

JEE MAIN REVISION MASTER: VECTOR ALGEBRA

2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Vector Algebra - Core Application	36 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Vector Algebra - Core Application (Very High Yield — 36 PYQs)

Core Principle: Key problem pattern in Kcet Vector Algebra.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value of λ for which the vectors $\vec{a}=2\hat{i}+\lambda\hat{j}+k\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+3k\hat{k}$ are orthogona..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\vec{a}=\hat{i}+\hat{j}+k\hat{k}, \vec{b}=\hat{j}-k\hat{k}$ and $\vec{a} \times \vec{c}=\vec{b}, \vec{a} \cdot \vec{c}=3$, then $\vec{c}$ i..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "If $\vec{a}=2\hat{i}+2\hat{j}-k\hat{k}, \vec{b}=\alpha\hat{i}+\beta\hat{j}+2k\hat{k}$ and $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$, then $\alpha+\beta=...$ " [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The value of λ for which the vectors $\text{vec}(\mathbf{a}) = 2\mathbf{i} + \lambda\mathbf{j} + \mathbf{k}$ and $\text{vec}(\mathbf{b}) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ are orthogonal is

- (A) $(5 / 2)$
- (B) $(-5 / 2)$ **[Correct]**
- (C) $(2 / 5)$
- (D) $(-2 / 5)$

Correct Answer: B

Solution Analysis: For two vectors to be orthogonal, their dot product must be zero. Given, $\text{vec}(\mathbf{a}) = 2\mathbf{i} + \lambda\mathbf{j} + \mathbf{k}$ and $\text{vec}(\mathbf{b}) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Now, $\text{vec}(\mathbf{a}) \cdot \text{vec}(\mathbf{b}) = 2(1) + \lambda(2) + 1(3) = 2 + 2\lambda + 3 = 5 + 2\lambda$. Since the vectors are orthogonal, $5 + 2\lambda = 0$
 $2\lambda = -5$ $\lambda = (-5 / 2)$ So, the correct option is Option B $(-5 / 2)$